Regression Diagnostics

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Outline

1. Regression Model and Its Assumptions
2. Model Diagnostics
3. Residual Analysis
4. Regression Performance Measures

Regression Model and Its Assumptions

The multiple linear regression model

- We observe n pairs $(\mathbf{x}^{(i)}, y^{(i)})$, $i = 1, \dots, n$, with p features in each $\mathbf{x}^{(i)}$
- The model, written for each observation:

$$y^{(i)} = \beta_0 + \beta_1 x_1^{(i)} + \cdots + \beta_p x_p^{(i)} + \varepsilon^{(i)}$$

- In matrix form, $\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\tilde{\mathbf{X}}$ is the $n \times (p+1)$ design matrix whose i-th row is $(1, x_1^{(i)}, \dots, x_p^{(i)})$
- The coefficients are estimated by least squares; inference (standard errors, intervals, tests) needs the assumptions on the next two slides

Assumptions of the MLR model (1/2): the mean and the features

- **Linearity**: the mean of y is linear in the coefficients, $\mathbb{E}[y^{(i)} \mid \mathbf{x}^{(i)}] = \beta_0 + \sum_{j=1}^{p} \beta_j x_j^{(i)}$
- **Zero-mean errors**: $\mathbb{E}[\varepsilon^{(i)} \mid \tilde{\mathbf{X}}] = 0$, so the errors are unrelated to the features
- **No perfect collinearity**: $\tilde{\mathbf{X}}$ has full column rank $p+1$, so $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ exists and the coefficients are uniquely determined

Assumptions of the MLR model (2/2): the errors

- **Uncorrelated errors**: $\mathrm{Cov}(\varepsilon^{(i)}, \varepsilon^{(k)}) = 0$ for $i \neq k$
- **Constant variance (homoscedasticity)**: $\mathrm{Var}(\varepsilon^{(i)}) = \sigma^2$ for every i
- **Normal errors**: $\varepsilon^{(i)} \sim \mathcal{N}(0, \sigma^2)$; needed for exact small-sample intervals and tests, not for the estimates themselves

These assumptions may not hold

- Real data often violate one or more of them: the true relation is curved, errors are correlated (over time or within units), the spread of the errors changes with the level of y, a few observations are unusual, or features are nearly collinear
- Consequences: wrong predictions (non-linearity); wrong standard errors, intervals and p-values (correlated errors, non-constant variance, non-normal errors); unstable coefficients (collinearity); a few points distorting the whole fit (outliers, high leverage)
- So after fitting, the assumptions have to be checked, not assumed

Model Diagnostics

What is model diagnostics?

- **Model diagnostics**: checks made after fitting to see whether the model assumptions hold, and whether the fit depends on a few observations or on redundant features
- The checks use quantities computed from the fit: residuals, studentized residuals, leverages and variance inflation factors
- Three questions: Is the linear model adequate? Which observations are unusual? Are some features redundant?
- Diagnostics examine whether the fit can be trusted; performance measures (later in this lecture) examine how well it fits

Coefficient estimates and fitted values

- Least squares chooses $\boldsymbol{\beta}$ to minimize $\mathrm{RSS} = \sum_{i=1}^{n} \left(y^{(i)} - \beta_0 - \sum_{j=1}^{p} \beta_j x_j^{(i)}\right)^2$, which gives

$$\hat{\boldsymbol{\beta}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$$

- The fitted values are obtained by plugging $\hat{\boldsymbol{\beta}}$ into the model:

$$\hat{\mathbf{y}} = \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}}, \qquad \hat{y}^{(i)} = \hat{\beta}_0 + \sum_{j=1}^{p} \hat{\beta}_j x_j^{(i)}$$

The hat matrix

- Substituting $\hat{\boldsymbol{\beta}}$ into $\hat{\mathbf{y}} = \tilde{\mathbf{X}}\hat{\boldsymbol{\beta}}$ expresses the fitted values as a linear map of the labels:

$$\hat{\mathbf{y}} = \tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}^\top\mathbf{y} = \mathbf{H}\mathbf{y}, \qquad \mathbf{H} = \tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}^\top$$

- $\mathbf{H}$ is the **hat matrix**: $n \times n$, symmetric and idempotent ($\mathbf{H}^2 = \mathbf{H}$), with $\operatorname{tr}(\mathbf{H}) = p + 1$
- $\mathbf{H}$ depends only on the features in $\tilde{\mathbf{X}}$, never on $\mathbf{y}$

The residual vector

- The residual for observation i is the gap between the observed and fitted value, $e^{(i)} = y^{(i)} - \hat{y}^{(i)}$; in vector form $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$
- Using $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, the residual vector is a linear function of $\mathbf{y}$:

$$\mathbf{e} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\,\mathbf{y}$$

- The residuals estimate the unobservable errors: $\varepsilon^{(i)}$ is never known, $e^{(i)}$ is computed from the data

The residual vector is a random vector

- $\mathbf{e}$ is a function of the random errors $\boldsymbol{\varepsilon}$, so each $e^{(i)}$ is a random variable and $\mathbf{e}$ is a random vector
- Its mean is $\mathbb{E}[\mathbf{e}] = \mathbf{0}$; its spread is described by the **variance–covariance matrix**

$$\mathrm{Cov}(\mathbf{e}) = \mathbb{E}\big[(\mathbf{e} - \mathbb{E}[\mathbf{e}])(\mathbf{e} - \mathbb{E}[\mathbf{e}])^\top\big] = \sigma^2(\mathbf{I} - \mathbf{H})$$

- $\mathrm{Cov}(\mathbf{e})$ is $n \times n$ and symmetric
- Diagonal entries are the individual variances: $\mathrm{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)})$
- Off-diagonal entries are the pairwise covariances: $\mathrm{Cov}(e^{(i)}, e^{(k)}) = -\sigma^2 H_{ik}$ for $i \neq k$
- The errors are uncorrelated, but the residuals generally are not, because $H_{ik} \neq 0$

The variance–covariance matrix of the residuals

- $\mathrm{Cov}(\mathbf{e}) = \sigma^2(\mathbf{I} - \mathbf{H})$ is an $n \times n$ symmetric matrix

$$\mathrm{Cov}(\mathbf{e}) = \sigma^2 \begin{pmatrix} 1 - h^{(1)} & -H_{12} & -H_{13} & \cdots & -H_{1n} \\ -H_{21} & 1 - h^{(2)} & -H_{23} & \cdots & -H_{2n} \\ -H_{31} & -H_{32} & 1 - h^{(3)} & \cdots & -H_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -H_{n1} & -H_{n2} & -H_{n3} & \cdots & 1 - h^{(n)} \end{pmatrix}$$

- Blue diagonal: the individual variances $\mathrm{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)})$
- Black off-diagonal: the pairwise covariances $\mathrm{Cov}(e^{(i)}, e^{(k)}) = -\sigma^2 H_{ik}$ for $i \neq k$

Where the hat matrix is used: leverage

- The diagonal of $\mathrm{Cov}(\mathbf{e}) = \sigma^2(\mathbf{I} - \mathbf{H})$ (previous slide) gives the variance of each residual:

$$\mathrm{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)}), \qquad h^{(i)} = H_{ii}$$

- $h^{(i)}$ is the **leverage** of observation i; it depends only on the features, and the leverages add up to $\sum_{i=1}^{n} h^{(i)} = p + 1$
- Even with constant error variance, residuals have unequal variances: high-leverage observations have smaller residual variance, because the fit is pulled towards them

Where the hat matrix is used: studentized residuals

- Raw residuals have unequal variances and carry the units of y; dividing by the estimated standard deviation fixes both:

$$r^{(i)} = \frac{e^{(i)}}{\hat{\sigma}\sqrt{1 - h^{(i)}}}, \qquad \hat{\sigma} = \mathrm{RSE}$$

- The leverage $h^{(i)} = H_{ii}$ comes from the hat matrix, so the studentized residual uses both the residual and the hat matrix
- Each $r^{(i)}$ has variance close to 1 and no units, so residuals can be compared across observations and across models
- With normal errors about 99.7% of the $r^{(i)}$ fall within ± 3: an observation with $|r^{(i)}| > 3$ is unusual, although among 400 observations one such value is expected by chance
- The studentized residual is also called the standardized residual; in R it is computed by `rstandard()`

Residual Analysis

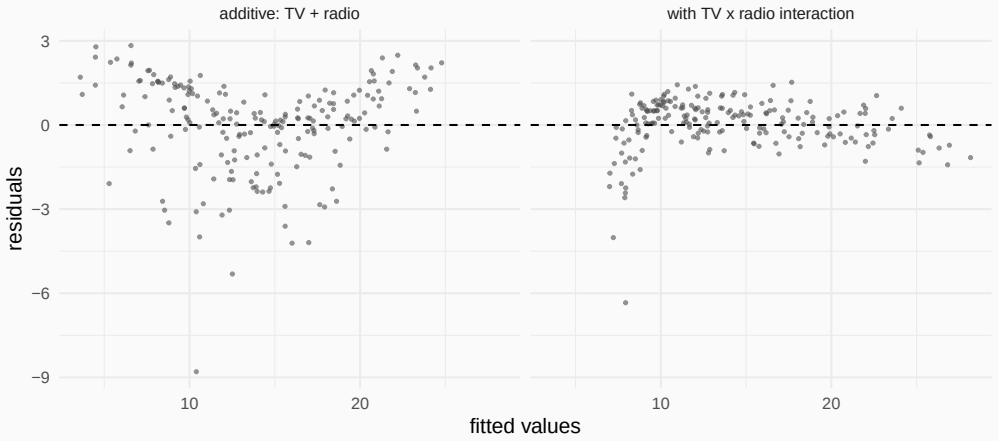
What is residual analysis?

- **Residual analysis**: examining the residuals $\mathbf{e}$ (or transformed versions of them) to judge whether the assumptions of the model hold
- If the assumptions hold, the residuals should look like random noise: centred at zero, with constant spread, no pattern against the fitted values, the features, or the observation order
- Main plots: residuals against fitted values, residuals against each feature, residuals against observation order, and the normal QQ plot of the residuals
- Observation-level quantities (leverage and studentized residuals) refine the check; they need the hat matrix, shown next

Problem 1: non-linearity, what to check

- **Plot**: residuals $e^{(i)}$ against fitted values $\hat{y}^{(i)}$, with a horizontal line at zero
- **If** the points follow a curve (U-shape or inverted U), **then** the linearity assumption is violated
- **If** the points scatter randomly around zero with no pattern, **then** linearity is plausible
- **Consequence**: predictions are biased in the ranges where the curve lies, and inference is unreliable
- **Remedies**: transform features ($\log x$, $\sqrt{x}$), add interaction terms, or fit a more flexible model

Problem 1: non-linearity, example



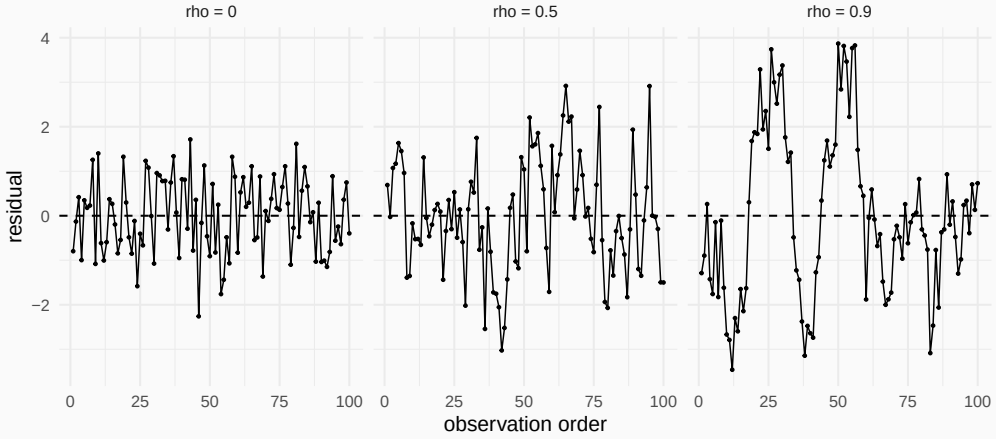
Problem 1: non-linearity, example (cont.)

- Dataset: Advertising, $n = 200$ markets (sales in thousands of units; TV and radio budgets in thousands of dollars)
- Left panel: additive model, sales $= \beta_0 + \beta_1$ TV $+ \beta_2$ radio $+ \varepsilon$; right panel: the same model with a TV $\times$ radio term
- Additive model: residuals are positive at the smallest and largest fitted values and negative in between, a U-shape that signals missing synergy between TV and radio
- With the interaction the curvature weakens and RSE falls from 1.68 to 0.94; a few large negative residuals remain

Problem 2: correlated errors, what to check

- **Plot**: residuals $e^{(t)}$ against observation order t, with a horizontal line at zero
- **If** neighbouring residuals are alike, with long runs of the same sign, **then** the uncorrelated-errors assumption is violated (successive errors are positively correlated)
- **If** the signs change at random from one observation to the next, **then** there is no evidence of correlation
- **Consequence**: standard errors are too small, so intervals are too narrow and p-values too small
- **Remedy**: time-series or mixed models, which are outside the scope of this lecture

Problem 2: correlated errors, example



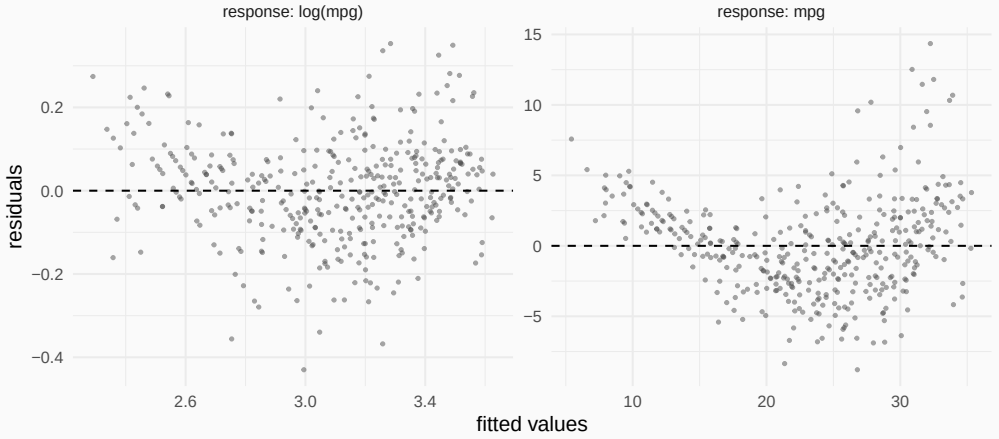
Problem 2: correlated errors, example (cont.)

- The above plots are from simulated datasets
- In the left plot, there is no autcorrelation among the errors, i.e., the observations are independent
- In the middple plot, autocorrelation is moderate (0.5). The residuals get stuck on one side of the horizontal axis
- In the right plot, autocorrelation is very high (0.9), the residuals get stuck on one side of the horizontal for much longer

Problem 3: non-constant variance, what to check

- **Plot**: residuals $e^{(i)}$ against fitted values $\widehat{y}^{(i)}$
- **If** the vertical spread grows or shrinks with the fitted value (a funnel), **then** the constant-variance assumption is violated
- **If** the vertical spread is about the same across the whole range, **then** constant variance is plausible
- **Consequence**: standard errors, intervals and tests become unreliable
- **Remedy 1**: transform y ($\log y$ or $\sqrt{y}$)
- **Remedy 2**: weighted least squares

Problem 3: non-constant variance, example



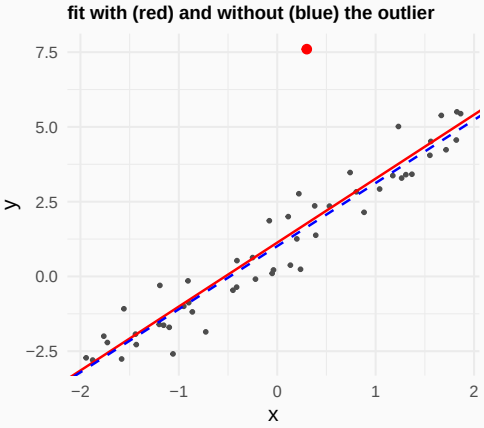
Problem 3: non-constant variance, example (cont.)

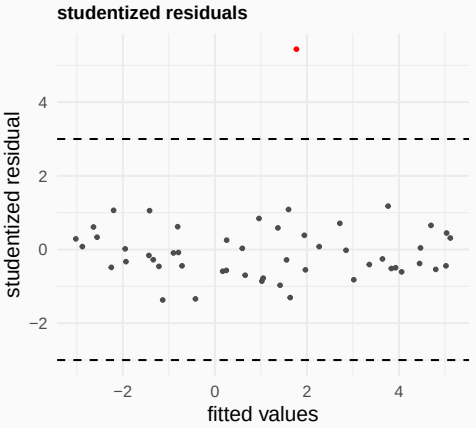
- Dataset and model: 'Auto' dataset, $n = 392$ cars;
  $\text{mpg} = \beta_0 + \beta_1 \text{ horsepower} + \beta_2 \text{ weight} + \beta_3 \text{ year} + \varepsilon$
- Left panel: $\log(\text{mpg})$ as response; right panel: mpg as response; the y-scales are free
- The mpg residuals fan out as the fitted value grows; their standard deviation grows from 2.35 to 2.48, 3.47 and 3.88 across the four quartiles of the fitted values
- For $\log(\text{mpg})$ the spread is nearly constant (0.116, 0.117, 0.130, 0.107), so the coefficients describe multiplicative effects

Problem 4: outliers, what to check

- **Plot**: studentized residuals $r^{(i)}$ against fitted values $\hat{y}^{(i)}$, with horizontal lines at ± 3
- **If** a point lies beyond ± 3, **then** it is a suspected outlier: its error is far outside the pattern shown by the rest of the data
- **Consequence**: an outlier inflates RSE and lowers R^2, and it can pull the fit towards itself
- **Action**: investigate the point (data-entry error or a genuine case) before deleting anything; refit and compare

Problem 4: outliers, example





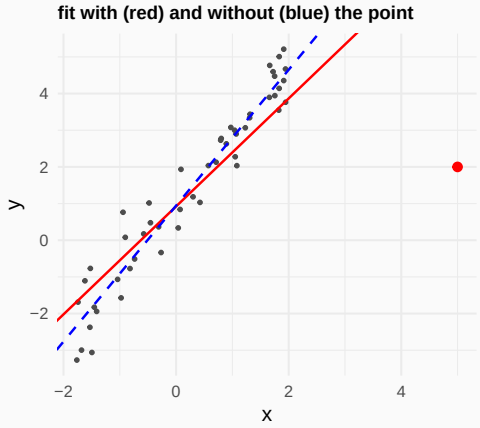
Problem 4: outliers, example (cont.)

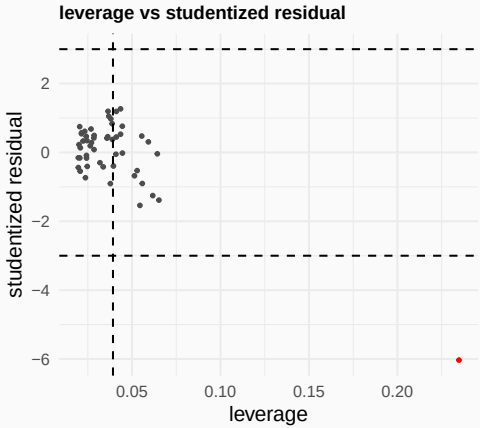
- Dataset (simulated): 50 points with $x \sim U(-2, 2)$ and $y = 1 + 2x + \varepsilon$, $\varepsilon \sim N(0, 0.7^2)$, plus one added point at $x = 0.3$, $y = 7.6$, six units above the line
- Left: fit with (red) and without (blue) the added point; right: studentized residuals against fitted values, with the outlier in red

Problem 5: high-leverage observations, what to check

- **Plot**: studentized residuals $r^{(i)}$ against leverage $h^{(i)}$, with a vertical line at the average $(p+1)/n$ and horizontal lines at ± 3
- **If** points lie far to the right of the average-leverage line, **then** they have high leverage: unusual feature values that can pull the fit towards themselves
- **If** a high-leverage point also has a large residual, **then** it is *influential*: removing it changes the fit a lot
- High leverage alone is not a violation of an assumption; the plot checks whether one point controls the fit
- **Remedy**: check the point, refit without it, and report both fits

Problem 5: high-leverage observations, example





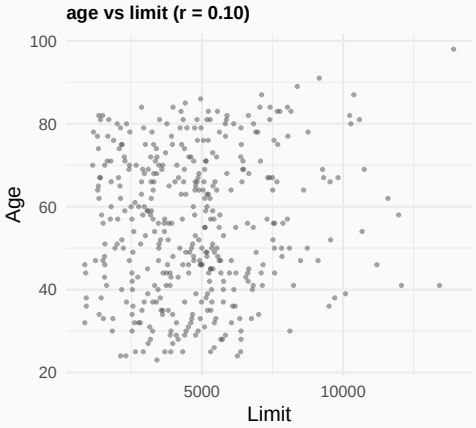
Problem 5: high-leverage observations, example (cont.)

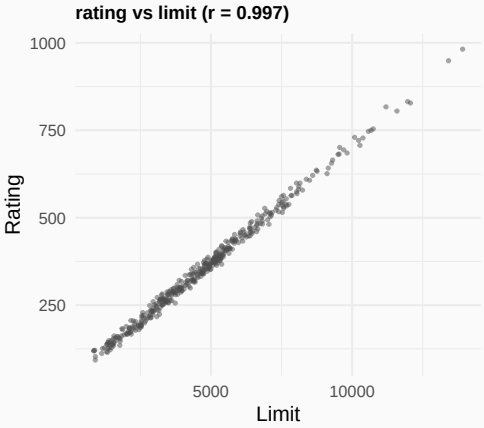
- Dataset (simulated): the 50 points of Problem 4 without the outlier, plus one added point at $x = 5$, $y = 2$; the model $y = \beta_0 + \beta_1 x + \varepsilon$ is fitted to all 51 points
- Left: fit with (red) and without (blue) the added point; right: studentized residuals against leverage
- The added point has the most extreme x, so its leverage is high; its studentized residual is also large and negative, so it is influential and the fitted line tilts visibly towards it
- Real data, Auto (model of Problem 3): the largest leverage is $h^{(i)} = 0.095$ (a 225-horsepower car), 9.3 times the average $4/392$

Problem 6: collinearity, what to check

- **Plot**: a scatterplot of each pair of features x_j against x_k, with the correlation r_{jk} in the title
- **If** the points lie almost on a straight line ($|r_{jk}|$ close to 1), **then** the two features carry nearly the same information, and the no-perfect-collinearity assumption is nearly violated
- Confirm with the VIF: values above 5 or 10 signal collinearity
- **Consequence**: unstable coefficients, inflated standard errors, and important features that look insignificant
- **Remedy**: drop one of the redundant features, or combine them

Problem 6: collinearity, example





Problem 6: collinearity, example (cont.)

- Dataset: Credit, $n = 400$ customers; response Balance, with features limit, rating and age
- Left: limit against age, $r = 0.10$, a diffuse cloud; right: limit against rating, $r = 0.997$, points lying almost on a line
- For Balance on age + limit, $\mathrm{SE}(\hat{\beta}_{\text{limit}}) = 0.0050$ ($p < 0.0001$); for rating + limit, 0.0638, 13 times larger, with $p = 0.70$

Detecting collinearity: variance inflation factor

- Pairwise correlations can miss collinearity among three or more features (multicollinearity); the **variance inflation factor** does not

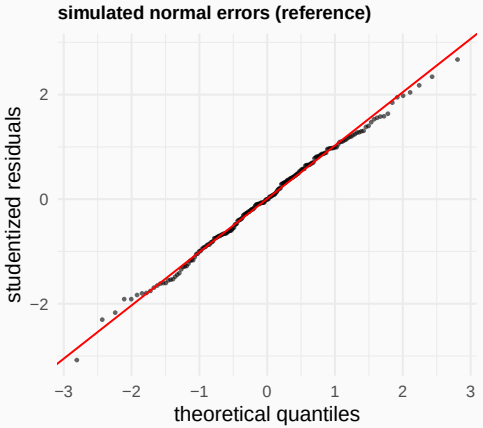
$$\text{VIF}(\hat{\beta}_j) = \frac{1}{1 - R^2_{x_j|x_{-j}}}$$

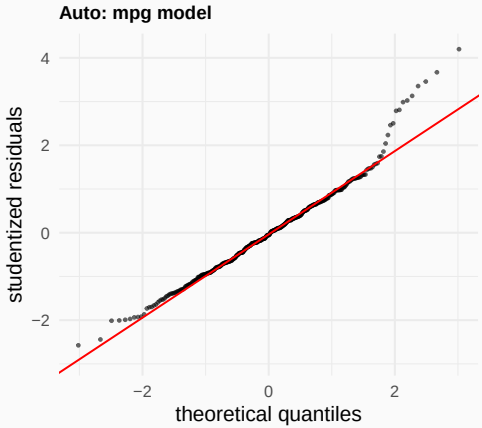
- $R^2_{x_j|x_{-j}}$ is the R^2 from regressing feature x_j on all the other features; VIF is at least 1, and values above 5 or 10 signal a problem
- Credit, age + rating + limit: VIF $= 1.01$ (age), 160.7 (rating), 160.6 (limit); with only age and limit: 1.01 and 1.01
- Remedies: drop one of the redundant features, or combine them (for example, average the standardized features)
- Predictions barely change ($R^2 = 0.750$ and 0.746 for the two models), but individual coefficients and their inference do

Problem 7: non-normal errors, what to check

- **Plot**: normal QQ plot of the studentized residuals $r^{(i)}$: sorted residuals against the quantiles of $\mathcal{N}(0, 1)$, with a reference line
- **If** the points follow the line, **then** normal errors are plausible
- **If** both ends bend away from the line, **then** the tails are heavier than normal: there are more extreme residuals than the normal model allows
- **If** only one end bends away, **then** the errors are skewed
- **Consequence**: exact small-sample intervals and tests are unreliable; with large n the central limit theorem keeps estimates and inference roughly valid
- **Remedies**: transform y, use robust regression, or bootstrap the inference

Problem 7: non-normal errors, example





Problem 7: non-normal errors, example (cont.)

- Left: simulated normal errors, $n = 200$; the points follow the line, which is what a normal QQ plot should look like
- Right: Auto, $n = 392$, model of Problem 3; the upper tail rises above the line, with the largest residual at 4.20 against about 3.0 expected under normality; the lower tail is less extreme than normal
- Skewness 0.67 and excess kurtosis 1.56: right-skewed with heavier tails; the largest residuals come from the high-mpg cars of Problem 4

Regression Performance Measures

Measures of the quality of fit

measure	definition	comment
MSE	$\mathrm{RSS}/(n-p-1)$	Squared units of y
RSE	$\sqrt{MSE}$	Units of y, estimates σ
MAE	$\frac{1}{n}\sum_{i=1}^{n}\left\|y^{(i)}-\hat{y}^{(i)}\right\|$	Units of y, less sensitive to outliers
R^2	$1-\mathrm{RSS}/\mathrm{TSS}$	Unit-free, never falls with more features
Adjusted R^2	$1-\frac{\mathrm{RSS}/(n-p-1)}{\mathrm{TSS}/(n-1)}$	Unit-free, penalizes each added feature

where $\mathrm{RSS}=\sum_{i=1}^{n}\left(y^{(i)}-\hat{y}^{(i)}\right)^2$.

- MSE and RSE punish large errors heavily (squares); MAE weighs all errors linearly, so a few outliers move it less

Interpreting R^2

- $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS} = \mathrm{ESS}/\mathrm{TSS}$ is also the squared correlation between $\mathbf{y}$ and $\hat{\mathbf{y}}$, and equals r^2 in simple regression
- It lies in $[0, 1]$ for a model with an intercept, and it never decreases when a feature is added: the larger model can always reproduce the smaller one by setting the new coefficient to zero
- It depends on the spread of the data: the same model fitted where x (and hence y) varies over a wider range has a higher R^2, and R^2 values of models with different responses (for example mpg and $\log$ mpg) are not comparable
- High R^2 does not mean the model is right (check the residuals); low R^2 does not mean it is useless, since real effects can be small relative to noise, for example in survey or behavioural data

Adjusted R-squared

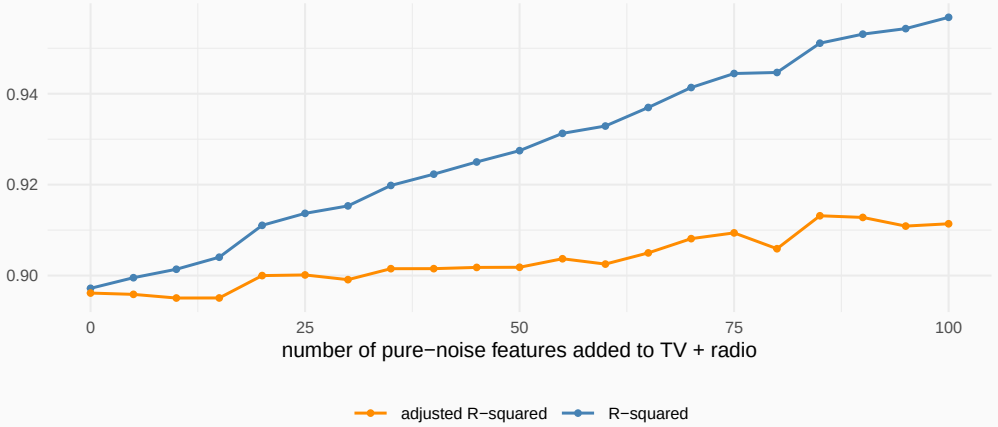
$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)} = 1 - (1-R^2)\,\frac{n-1}{n-p-1}$$

- It compares two unbiased variance estimates: the residual variance $\hat{\sigma}^2$ and the variance of y; the factor $(n-1)/(n-p-1)$ grows with p and penalizes each added feature
- Adding a feature raises R^2_{adj} exactly when the feature's $|t|$-statistic exceeds 1; a weak bar, but it stops R^2_{adj} from rising for pure noise
- Advertising: adding newspaper has $|t| = 0.18 < 1$, so R^2_{adj} falls from 0.8962 to 0.8956 while R^2 stays 0.8972

Adjusted R-squared with useless features

- Dataset: Advertising, $n = 200$; model: sales on TV + radio, with pure-noise features added to it, drawn independently of sales
- Plot: R^2 and R^2_{adj} against the number of noise features added cumulatively, in steps of 5 up to 100
- R^2 can only rise, here from 0.897 to about 0.95 with 100 noise features; R^2_{adj} stays near 0.90 and shows no systematic gain

Adjusted R-squared with useless features (cont.)



Comparing regression models on the Advertising data

model	RSE	RMSE	MAE	R^2	adjusted R^2
TV	3.259	3.242	2.550	0.6119	0.6099
TV + radio	1.681	1.669	1.254	0.8972	0.8962
TV + radio + newspaper	1.686	1.669	1.252	0.8972	0.8956
TV $\times$ radio	0.944	0.934	0.660	0.9678	0.9673

- RMSE and MAE are in thousands of units of sales: the typical error falls from about 3.2 (TV only) to about 0.9 (TV $\times$ radio)
- Adding newspaper leaves R^2 unchanged (0.8972) but lowers adjusted R^2 (0.8962 to 0.8956) and raises RSE: the penalty exposes a useless feature
- All numbers here are computed on the training data used to fit the models

Summary

- **Assumptions**: linearity, zero-mean errors, full column rank, uncorrelated errors, constant variance and (for exact inference) normal errors; these may not hold, so they must be checked
- **Residuals**: $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$, with $\mathrm{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)})$; studentized residuals $r^{(i)} = e^{(i)}/(\hat{\sigma}\sqrt{1 - h^{(i)}})$
- **Diagnostics**: residual plots reveal non-linearity, correlated errors and non-constant variance; studentized residuals flag outliers, leverage flags unusual feature values, VIF flags collinearity
- **Performance**: MSE, RMSE, MAE, RSE, R^2 and adjusted R^2 measure the fit to the training data; RSE and adjusted R^2 account for the degrees of freedom; training error keeps falling as features are added, test error does not

Questions?